

Review

Polar Coordinates: $x = r \cos \theta$; $r = \sqrt{x^2 + y^2}$
 $y = r \sin \theta$; $\theta = \arctan(y/x)$

Derivative matrices:

$$D = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}; \quad D^{-1} = \begin{bmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\frac{1}{r} \sin \theta & \frac{1}{r} \cos \theta \end{bmatrix}$$

Jacobian: $\frac{\partial(x,y)}{\partial(r,\theta)} = |J(r,\theta)| = |\det D| = r$; $\frac{\partial(r,\theta)}{\partial(x,y)} = |J(x,y)| = |\det D^{-1}| = \frac{1}{r}$

Lamé Coefficients:

$$h_1 = \left\| \frac{\partial \mathbf{x}}{\partial r} \right\| = 1, \quad h_2 = \left\| \frac{\partial \mathbf{x}}{\partial \theta} \right\| = r$$

Frames:

$$\begin{aligned} \hat{e}_r &= \cos \theta \hat{i} + \sin \theta \hat{j} \\ \hat{e}_\theta &= -\sin \theta \hat{i} + \cos \theta \hat{j} \end{aligned}$$

$$\begin{aligned} \hat{i} &= \cos \theta \hat{e}_r - \sin \theta \hat{e}_\theta \\ \hat{j} &= \sin \theta \hat{e}_r + \cos \theta \hat{e}_\theta \end{aligned}$$

Vector Calculus:

$$\vec{F} = F_1 \hat{i} + F_2 \hat{j} = F_r \hat{e}_r + F_\theta \hat{e}_\theta, \text{ where } F_r = F_1 \cos \theta + F_2 \sin \theta$$

$$F_\theta = -F_1 \sin \theta + F_2 \cos \theta$$

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} = \frac{\partial f}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{e}_\theta$$

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} = \frac{1}{r} \frac{\partial}{\partial r} (r F_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (F_\theta)$$

$$\iint_{R(x,y)} f \, dx \, dy = \iint_{R(r,\theta)} f \frac{\partial(x,y)}{\partial(r,\theta)} \, dr \, d\theta = \iint_{R(r,\theta)} f \, r \, dr \, d\theta$$